

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE; Semester Final Examination; Spring-2023

Course Code: MATH-2407

Course Title: Mathematics-IV

Time: 2:30 hours

Total Marks: 50

Answer all the questions. The figures in the right-hand margin indicate full marks.

Group A

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles, also find the Fourier series for the function. 10 CO3 U

$$y = f(t) = \begin{cases} 0; & -3 \leq t < 0 \\ 3; & 0 \leq t < 3 \end{cases}$$

- Or Derive the complex form of the Fourier series. 10 CO3 U

- 2.a Find Harmonic analysis of the given Fourier series 5 CO4 An

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+2}}{n} \sin \frac{n\pi t}{6}$$

- 2.b Find the Convolution Sum $x[n] * h[n]$ for the following functions 5 CO4 An

$$\begin{aligned} x[n] &= 3; n=0 \\ &= 1; n=1 \end{aligned} \quad \text{and} \quad \begin{aligned} h[n] &= 1; n=0 \\ &= 3; n=1 \end{aligned}$$

Where n is the time index

Group B

- 3.a Solve the following Initial Value Problem (IVP) by Laplace Transform: 5 CO3 Ap
 $Y'' + Y = t, Y(0) = 1, Y'(0) = 2$

- Or Find Laplace Transform $t^2 \sin 2t$ using the Multiplication theorem. 5 CO3 Ap

- 3b Sketch the waveforms for the following signals 5 CO1 Ap
 $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

- Or $x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$ 5 CO1 Ap

- 4.a State first shift theorem. Using the theorem evaluate $\mathcal{L}\{e^{-4t}t^2\}$ 5 CO1 U

- 4.b Define unit step function. Sketch the following function, 5 CO1 U
 $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

- 5.a Write MATLAB code to sketch line spectrum (at least 6) for the following 4 CO5 Ap

$$\text{Fourier series } \underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2}_{\text{DC value}} + \underbrace{\left[\sum_{n=1}^{\infty} (\cos 2n\pi + 1) \sin \frac{n\pi t}{6} \right]}_{\text{AC value}}$$

- 5.b Make a function in MATLAB environment to raise a complex wave $f(t)$ in the time interval of $[-4, 20]$ for the following Fourier series: 6 CO5 Ap

$$f(t) = 4\pi + \sum_{n=1}^{\infty} \frac{4}{n\pi} \cos 2n\pi t$$

OR

$S \rightarrow aXbX$
 $X \rightarrow aY \mid bY \mid \epsilon$
 $Y \rightarrow X \mid c$

- b) Find out whether the following problem is decidable or not.
 Is a number 'm' prime?

3 CO4 E

OR

Determining if a given graph G is connected.

- c) What are NP-complete and NP-Hard problems? How can you show that a problem is NP-complete?

3 CO5 U

International Islamic University Chittagong
 Department of Computer Science & Engineering
 B.Sc. in CSE Final Examination, Autumn 2022

Course Title: Mathematics-IV Course Code: MATH-2407 (New)
 Course Title: Mathematics-V Course Code: MATH-3501 (Old)

Time: 2 Hours 30 Minutes

Full Marks: 50

[The figures in the right-hand margin indicate full marks, Separate answer scripts must be used for Group A and Group B, Course Outcomes and Bloom's Levels are mentioned in additional Columns]

Course Outcomes (COs) of the Questions	
CO1	Demonstrate the understanding of the basic principles and operations on set theory, mathematical operations with complex numbers, geometrical interpretation and the related fundamental theories involving complex functions and the concept of transformation in a complex plane and waves
CO2	Apply the concept of transformation of an object into complex space and different mathematical operation of complex functions
CO3	Use Fourier series, Laplace's Transforms, Fourier Transform in different scenario
CO4	Analyze the harmonics & spectrum of different types of waves
CO5	Demonstrate the harmonic analysis using MATLAB

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group A

Answer the following questions

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles, CO3 U

$$y = f(t) = \begin{cases} 0; & -4 \leq t < 0 \\ 4; & 0 \leq t < 4 \end{cases}$$

Also find the Fourier series for the function.

OR

Derive the complex form of Fourier series.

2. a) Find Harmonic analysis of the given Fourier series CO4 An

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+2}}{n} \sin \frac{n\pi t}{3}$$

- b) Plot the line (at least 6) spectrum (discrete frequency spectra) for the Fourier series CO4 An

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{5}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{4}{n} \sin 2n\pi t}_{\text{AC value}}$$

Group A
Answer the following questions

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles, CO3 U

$$y = f(t) = \begin{cases} 0; & -4 \leq t < 0 \\ 4; & 0 \leq t < 4 \end{cases}$$

Also find the Fourier series for the function.

OR

Derive the complex form of Fourier series.

2. a) Find Harmonic analysis of the given Fourier series CO4 An

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi t}{3}$$

- b) Plot the line (at least 6) spectrum (discrete frequency spectra) for the Fourier series CO4 Ar

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Group B
Answer the following questions

3. a) Prove that $L\{f'(t)\} = s^2 L\{f(t)\} - s f(0) - f'(0)$ CO3 Ap 5

OR

Find the inverse Laplace transform of $\frac{s+4}{s(s-1)(s-2)}$ U

- b) Express the following function in terms of unit step functions and find its Laplace transform CO3 Ap 5

$$f(t) = \begin{cases} 10; & t < 3 \\ 8; & t > 3 \end{cases}$$

OR

Find Fourier Transform of

$$f(t) = \begin{cases} 1 & ; 0 \leq t < 1 \\ -1 & ; -1 \leq t < 0 \\ 0 & ; |t| > 1 \end{cases}$$

4. a) State first shift theorem. Using the theorem evaluate $L\{e^{-4t}t^2\}$ CO1 U

- b) Define unit step function. Sketch the following function, CO1 U

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

5. a) Write MATLAB code to sketch line spectrum (at least 6) for the following Fourier series CO5 Ap

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2}_{\text{DC value}} + \underbrace{\left[\sum_{n=1}^{\infty} (\cos n\pi + 1) \sin \frac{n\pi t}{3} \right]}_{\text{AC value}}$$

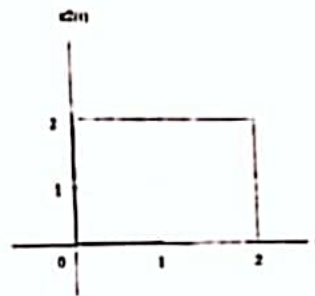
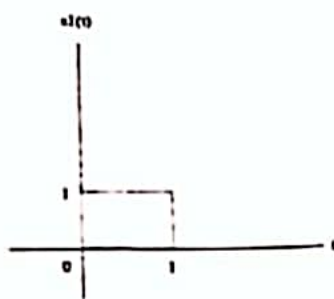
- b) Make a function in MATLAB environment to raise a complex wave $f(t)$ in the time interval of $[-4, 20]$ for the following Fourier series: CO5 Ap

$$f(t) = 4\pi + \sum_{n=1}^{\infty} \frac{3}{n\pi} \cos n\pi t$$

- c) If $x[n] = 5; n=0$ and $h[n] = 3; n=0$ CO5 Ap
 $= 6; n=1$ $= -2; n=1$

Write MATLAB code to find the convolution sum of the above signals.

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Group B
Answer the following questions

~~5+4=2~~

3. Find the Fourier transform of the function $f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & -1 < t < 0 \\ 0, & \text{otherwise} \end{cases}$

CO3 App 5

- b) Find the inverse Laplace transform of $\frac{s+4}{s(s-1)(s-2)}$

CO3 App 5

Or:

CO3 App 5

- b) Find $L[u(t-3)] = \frac{e^{-3s}}{s}$

4. a) Solve the Initial Value Problem (IVP) by Laplace Transformation.

CO3 App 5

$$Y'' - 3Y' + 2Y = 4e^{2t}, \quad Y(0) = -3, \quad Y'(0) = 5$$

Or

- b) Solve the following Initial Value Problem (IVP) by Laplace Transform:

CO3 App 5

$$Y'' + Y = 1, \quad Y(0) = 1, \quad Y'(0) = -2$$

- c) Define unit step function. Sketch the waveforms for the following signals

CO1 App 5

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

5. Write MATLAB code to sketch line spectrum (at least 6) for the following Fourier series

CO5 App 5

$$f(t) = \underbrace{2.5}_{\text{Complex wave}} + \underbrace{\left[-\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4}\right]}_{\text{AC value}}$$

- b) Make a function in MATLAB environment to raise a complex wave $f(t)$ in the time interval of $[-4, 20]$ for the following Fourier series:

CO5 App

$$f(t) = \pi - \sum_{n=1}^{\infty} \frac{2}{n} \sin n\pi$$

- If $x[n] = 1; n=0$ and $n[n] = 2; n=0$
 $= -1; n=1$ $= -2; n=1$

Write MATLAB code to find the convolution sum of the above signals.

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Time: 2 Hours 30 Minutes

Full Marks: 50

- (i) The figures in the right-hand margin indicate full marks
 (ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Letter Symbols Meaning	Bloom's Levels of the Questions					
	R	U	App	An	E	C
	Remember	Understand	Apply	Analyze	Evaluate	Create

Group A

Answer the following questions

1. Define Fourier series in the interval $(-L, L)$. Sketch the following function for four cycles. CO3 App 10

$$y = f(t) = \begin{cases} 0; & -4 \leq t < 0 \\ 4; & 0 \leq t < 4 \end{cases}$$

Also find the Fourier series for the function

2. a) Find the Harmonic Analysis (at least 4) for the given Fourier series. CO4 An 5

$$f(t) = \frac{5}{2} - \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4}$$

Or:

2. a) Plot the line (at least 6) spectra (discrete frequency spectra) for the Fourier series CO4 An

$$\underline{f(t)} = \underline{\frac{5}{2}} + 2 \sum_{n=1}^{\infty} \frac{1}{n} \sin n\pi$$

2. b) Find Convolution Sum $x[n] * h[n]$ for the following functions

$$x[n] = \begin{cases} 1; & n=0 \\ -1; & n=1 \end{cases}$$

and

$$h[n] = \begin{cases} 1; & n=0 \\ 1; & n=1 \end{cases}$$

Where n is the time index

2. b) Find convolution integral of $x(t) * x_2(t)$

10 / 13 CO4 An

CO4 An

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